PROJECT REPORT
ON

AI-BASED DRONE DETECTION, TRACKING AND NEUTRALISATION PROTOTYPE USING YOLOv8 AND ARDUINO

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE
OF

BACHELOR OF TECHNOLOGY
IN
MECHANICAL ENGINEERING

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CERTIFICATE

This is to certify that the Project titled "**AI-BASED DRONE DETECTION, TRACKING AND NEUTRALISATION PROTOTYPE USING YOLOv8 AND ARDUINO"** submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Mechanical Engineering by **"Vikas (22BME012), Afaaf Siddiqui (22BME017), Amaan Hasan Farooqi (22BME018) and Mohammad Sahil Khan (22BME025)"** is a bonafide record of the candidate's own work carried out by them under my supervision and guidance.

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We would like to take this opportunity to express our sincere indebtedness and sense of gratitude to all those who have contributed greatly towards the successful completion of this Project.

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DECLARATION

We hereby declare that the thesis entitled “AI-Based Vision-Guided Drone Detection, Tracking and Missile Launcher Actuation System using YOLOv8 and Arduino” submitted to the Department of Mechanical Engineering, Faculty of Engineering and Technology, Jamia Millia Islamia, in partial fulfillment of the requirements for the degree of Bachelor of Technology (B.Tech), is our original work carried out under the supervision of Dr. Mohd Suhaib.

We further declare that this work has not been submitted previously, either in full or in part, for the award of any degree, diploma, or fellowship at any other institution or university. All the information, data, references, and materials used in this thesis have been properly acknowledged.

We certify that the work presented in this thesis is genuine and has been completed in accordance with academic ethics and integrity guidelines prescribed by the university.

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ABSTRACT

The rapid increase in the use of Unmanned Aerial Vehicles (UAVs), commonly known as drones, has introduced significant challenges related to security, surveillance, and public safety. Unauthorized drones entering restricted areas such as airports, military zones, and industrial facilities may pose serious risks due to their ability to carry cameras, sensors, or harmful payloads. Traditional surveillance systems often face limitations in detecting small aerial targets because of their compact size, fast movement, and varying environmental conditions. Therefore, the development of intelligent and automated drone detection systems has become an important area of research.

The project presents an AI-Based Drone Detection, Tracking and spark-actuated missile launcher Prototype using YOLOv8 and Arduino. A 1080p camera is used to capture live video feeds, which are processed using the YOLOv8 deep learning algorithm trained on a custom drone dataset collected through Roboflow. The trained model detects and tracks the drone in real time by generating bounding boxes and positional coordinates.

The positional coordinates are transmitted to an Arduino Uno microcontroller through serial communication. The Arduino controls a pan-tilt mechanism using MG996R servo motors to continuously align the camera and launcher system with the moving target. AA spark-generation mechanism is used to actuate the missile launcher during target engagement. The launched missile neutralizes the drone target. The system also incorporates target alignment, lead angle estimation, distance approximation, and proportional servo control for accurate target tracking.

The proposed prototype demonstrates the successful integration of deep learning, visual servoing, missile-launch actuation mechanisms, and embedded automation technologies for intelligent surveillance applications. The system provides advantages such as real-time detection capability, low implementation cost, efficient target tracking, safe operation, and easy scalability for future improvements. The developed prototype provides a strong foundation for future research in AI-based surveillance, autonomous tracking systems, and intelligent drone interception technologies.

Keywords: YOLOv8, Drone Detection, Computer Vision, Arduino Uno, Visual Servoing, Spark-generation module, Object Tracking, Deep Learning, Surveillance System.

PROBLEM STATEMENT

The rapid growth in the use of Unmanned Aerial Vehicles (UAVs), commonly known as drones, has created major challenges in surveillance, public safety, and security. Although drones are widely used in applications such as aerial photography, agriculture, mapping, and delivery systems, their misuse in restricted areas has become a serious concern. Unauthorized drones entering airports, military zones, industrial plants, and public areas may pose threats through illegal surveillance, smuggling, or disruption of critical operations.

Traditional surveillance systems often face difficulties in detecting and tracking small aerial objects due to their compact size, high speed, and unpredictable movement. Conventional methods such as radar and manual monitoring may be expensive, less efficient, and prone to human error.

Recent advancements in Artificial Intelligence (AI), Deep Learning, and Computer Vision technologies provide effective solutions for real-time drone detection and tracking. Therefore, there is a growing need for an intelligent, low-cost, and automated surveillance system capable of detecting and tracking drones accurately in real time.

The proposed project aims to develop an AI-based drone detection and tracking system using YOLOv8, OpenCV, Arduino Uno, and servo-based automation. The system is designed to detect drones through live video streams, track their movement using a pan-tilt mechanism, and demonstrate intelligent real-time surveillance under controlled laboratory conditions.

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CHAPTER 1
INTRODUCTION

1.1 Introduction to Drone Technology

Unmanned Aerial Vehicles (UAVs), commonly known as drones, are aircraft systems that operate without an onboard human pilot. These systems are controlled either remotely by operators or autonomously through embedded software and sensors. In recent years, drone technology has experienced rapid growth due to advancements in artificial intelligence, wireless communication, embedded systems, and lightweight materials.

Drones are now widely used in various civilian and military applications such as aerial photography, surveillance, agriculture monitoring, package delivery, disaster management, mapping, and environmental monitoring. Their ability to access difficult terrains and capture real-time data has made them highly beneficial in modern industries.

However, the increasing accessibility and affordability of drones have also introduced several security threats. Unauthorized drones may enter restricted areas such as airports, military zones, industrial plants, and public events, causing serious safety concerns. Therefore, there is an increasing demand for intelligent systems capable of detecting and tracking drones in real time.

Figure 1.1: Autonomous Drone

1.2 Need for Drone Detection Systems

The rapid rise in drone usage has created new challenges for security and surveillance agencies. Traditional surveillance systems often fail to identify small drones due to their compact size, fast movement, and varying environmental conditions.

Drone detection systems are essential for preventing unauthorized aerial surveillance, protecting sensitive locations, monitoring restricted airspaces, enhancing public safety, and reducing risks of illegal activities.

Modern anti-drone systems use technologies such as radar, radio frequency analysis, acoustic sensors, thermal imaging, and computer vision. Among these, computer vision-based systems are becoming increasingly popular because they are cost-effective, flexible, and capable of real-time object detection.

The proposed project focuses on a computer vision-based drone detection and tracking system using artificial intelligence and embedded hardware integration.

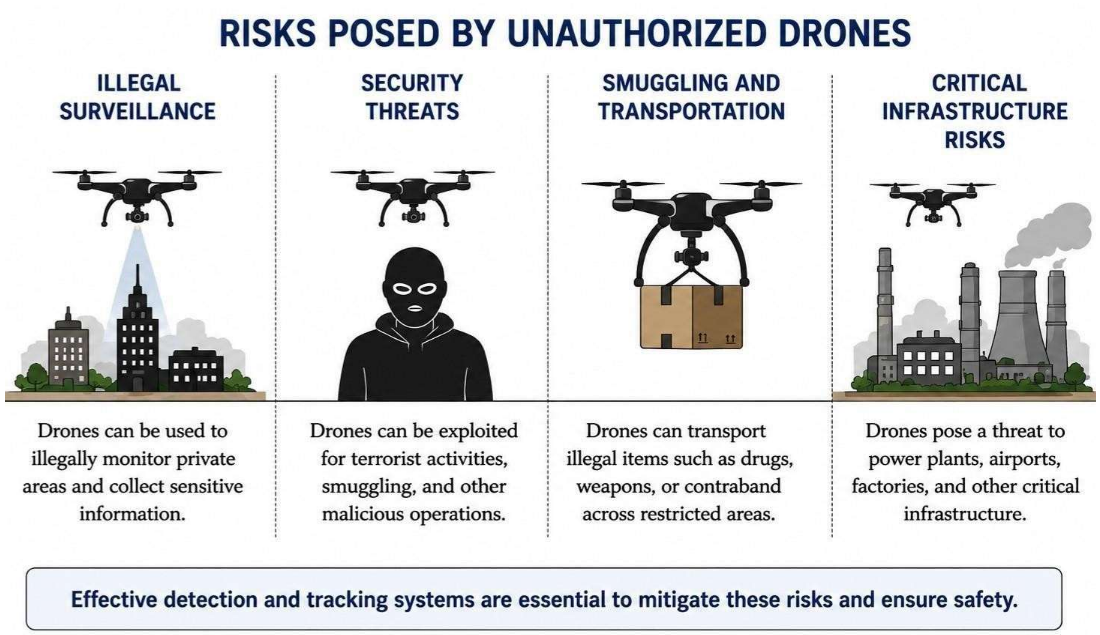

Figure 1.2 : Risks posed by drones

1.3 Artificial Intelligence in Surveillance Systems

Artificial Intelligence (AI) is one of the most important technologies used in modern surveillance and security systems. AI enables machines and computer systems to perform tasks that normally require human intelligence, such as learning, decision-making, image recognition, pattern analysis, and object identification. In surveillance systems, AI helps in automating monitoring operations and improving the efficiency and accuracy of security applications.

Traditional surveillance systems mainly depend on human operators to continuously observe video feeds and identify suspicious activities. However, human monitoring becomes difficult and inefficient when dealing with multiple cameras or long-duration surveillance. Artificial intelligence overcomes these limitations by enabling automatic analysis of visual information in real time.

Artificial intelligence-based surveillance systems use machine learning and deep learning algorithms to analyze images and video streams. Deep learning algorithms, especially Convolutional Neural Networks (CNNs), are highly effective for image processing and object detection tasks.

In this project, artificial intelligence is used for real-time drone detection using the YOLOv8 deep learning algorithm. The system continuously processes video frames captured by the camera and identifies drone targets automatically. The use of AI improves detection speed, accuracy, and automation capabilities in modern surveillance systems.

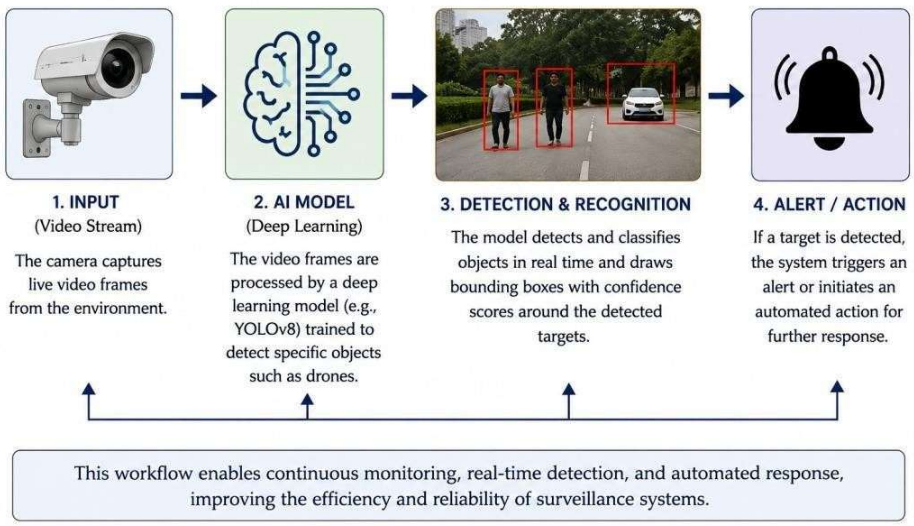

Figure 1.3: AI Workflow

1.4 Introduction to YOLOv8

YOLO (You Only Look Once) is a state-of-the-art object detection algorithm widely used in computer vision applications. YOLO is designed for real-time object detection where both speed and accuracy are important.

YOLOv8, developed by Ultralytics, is the latest version of the YOLO family and provides improved accuracy, faster processing speed, and better small-object detection capabilities. The algorithm processes the complete image in a single pass to predict object locations and classifications simultaneously.

YOLOv8 is especially suitable for drone detection because drones often appear as small moving targets within video frames. The algorithm is capable of detecting drones efficiently under different lighting and environmental conditions.

In this project, YOLOv8 is trained using a custom drone dataset collected from Roboflow. The trained model processes live webcam video streams and generates bounding boxes and positional coordinates for tracking purposes.

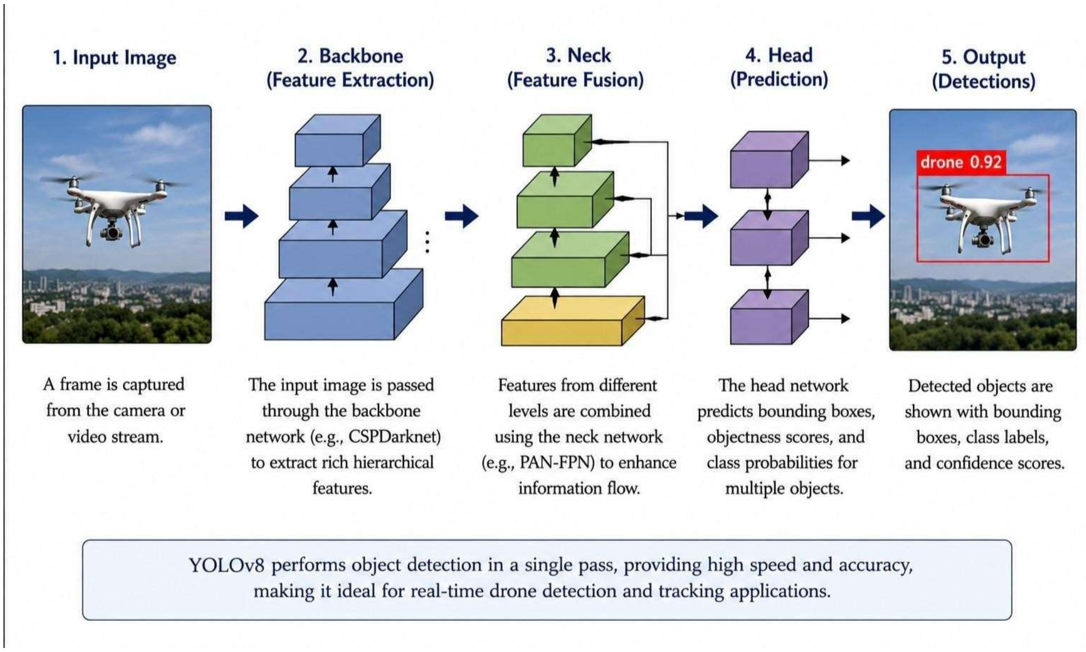

Figure 1.4: YOLO Object Detection Workflow

1.5 Arduino-Based Automation System

Arduino is an open-source microcontroller platform widely used for automation and embedded system applications. It provides an easy interface for controlling hardware components such as motors, sensors, relays, and actuators.

In this project, the Arduino Uno acts as the hardware controller responsible for controlling pan and tilt servo motors, and receiving tracking coordinates from Python. It also performs SCAN mode, TRACK mode, TARGET LOCK verification, and FIRE command generation for spark-actuated missile launcher activation using a transistor-based switching circuit.

Servo motors are used to rotate the camera and launcher system in horizontal and vertical directions, enabling automatic target tracking. Communication between the Python program and Arduino is established through serial communication using a USB connection.

The integration of artificial intelligence with Arduino-based hardware creates a low-cost intelligent tracking system suitable for educational and research purposes.

1.6 Computer Vision and Object Tracking

Computer vision is a branch of artificial intelligence that enables machines to analyze and interpret visual information obtained from images and video streams. Computer vision systems use image processing, machine learning, and deep learning algorithms to identify objects and analyze their movement.

Object detection refers to identifying and locating objects present within an image or video frame. In this project, YOLOv8 is used to detect drones from live video feeds captured using a webcam.

Object tracking refers to continuously following the movement of a detected object across multiple video frames. Once the drone is detected, the tracking system continuously updates the target position and adjusts the camera and launcher orientation accordingly.

The proposed system uses a pan-tilt mechanism mounted with a camera and spark-actuated missile launcher module. Servo motors rotate the mechanism horizontally and vertically so that the detected drone remains centered within the camera view. A coordinate-based visual-servo control mechanism is implemented where the positional error between the drone center coordinates and image center coordinates is continuously minimized for accurate target tracking and TARGET LOCK verification. The Arduino controller generates LEFT, RIGHT, UP, and DOWN movement commands for real-time servo alignment and activates the spark-actuated missile launcher mechanism through transistor-based switching after FIRE command verification.

Computer vision-based tracking systems provide advantages such as real-time operation, accurate target localization, low-cost implementation, and high automation capability.

1.7 Objectives of the Project

The primary objectives of this project are:

1. To develop a real-time drone detection system using YOLOv8.
2. To train a deep learning model using drone image datasets.
3. To integrate computer vision with Arduino-based hardware control.
4. To design a pan-tilt tracking mechanism using servo motors.
5. To automate target tracking using real-time positional data.
6. To implement TARGET LOCK verification and Arduino-controlled spark-actuated missile launcher activation using embedded automation techniques.

1.8 Scope of the Project

The scope of this project includes real-time drone detection using YOLOv8, webcam-based surveillance, coordinate-based servo tracking, Arduino hardware integration, SCAN and TRACK operations, TARGET LOCK verification, spark-actuated missile launcher activation, and experimental testing under laboratory conditions.

The system is designed primarily for academic demonstration and research purposes.

It provides a foundation for future advancements in intelligent drone surveillance, embedded automation, AI-based tracking systems, and real-time response mechanisms.

1.9 Advantages of the Proposed System

The proposed system offers several advantages such as real-time drone detection and tracking, low-cost implementation, AI-based automation, SCAN and TRACK operations, TARGET LOCK verification, Arduino-controlled spark-actuated missile launcher activation, efficient YOLOv8-based object detection, and easy hardware-software integration.

The project demonstrates the practical integration of artificial intelligence, computer vision, embedded systems, coordinate-based tracking, and spark-actuated missile launcher activation technologies in intelligent drone surveillance systems.

1.10 Applications of the System

The developed system can be applied in border security, military surveillance, airport monitoring, industrial safety, smart city surveillance, restricted area monitoring, and research and educational demonstrations.

1.11 Conclusion

This chapter introduced drone technology, the need for intelligent drone surveillance systems, artificial intelligence-based detection techniques, and the role of YOLOv8 in real-time drone detection and tracking. The integration of computer vision, Arduino-based automation, coordinate-based servo tracking, TARGET LOCK verification, and spark-actuated missile launcher activation has also been discussed. The next chapter presents a detailed review of existing literature and related research work in the field of drone detection and tracking systems.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

The literature review is an important part of any research project as it provides an understanding of the existing technologies, methodologies, and research developments related to the proposed work. In recent years, drone detection and tracking systems have become an active area of research due to the increasing use of Unmanned Aerial Vehicles (UAVs) in both civilian and military sectors.

Researchers across the world have developed various techniques for drone detection using radar systems, thermal imaging, radio frequency analysis, acoustic sensors, and computer vision approaches. Among these methods, artificial intelligence and deep learning-based computer vision techniques have shown significant improvements in real-time object detection and tracking applications.

This chapter presents a review of previous research work related to drone detection systems, YOLO-based object detection methods, embedded automation systems, and intelligent surveillance technologies.

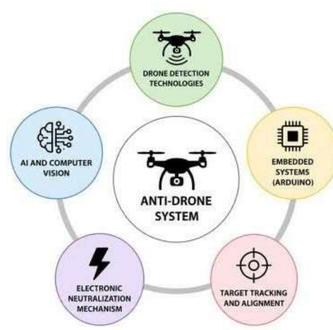

Figure 2.1: AI-Based Anti-Drone Systems

2.2 Drone Detection Technologies

Drone detection technologies are designed to identify and monitor unauthorized drones entering restricted airspaces. Different detection methods are available depending on the operational requirements, environmental conditions, and system complexity.

Radar-based systems are widely used in military applications because they can detect long-range aerial objects. However, radar systems are expensive and often struggle to detect small drones with low radar cross-sections.

Acoustic-based systems use microphones and sound analysis algorithms to recognize the unique sound signatures generated by drone propellers. These systems are cost-effective but their performance decreases in noisy environments.

Radio Frequency (RF) detection systems identify communication signals between drones and their controllers. Although RF systems are effective, they may fail when drones operate autonomously without communication links.

Thermal imaging systems detect heat signatures produced by drones, especially during nighttime operations. However, thermal cameras are expensive and computationally intensive.

Computer vision-based systems use cameras and artificial intelligence algorithms for object detection and tracking. These systems are low-cost, flexible, and suitable for real-time surveillance applications. Due to these advantages, computer vision has been selected for the proposed project.

Table 2.1 Comparison of Object Detection Algorithms

Algorithm	Speed	Accuracy	Real-Time Capability	Application
YOLOv8	Very High	High	Excellent	Drone Detection
Faster R-CNN	Medium	Very High	Limited	Research Applications
SSD	High	Medium	Good	Mobile Vision Systems
Haar Cascade	Very High	Low	Good	Basic Object Detection

Table 2.2 Comparison of Counter-UAV Technologies

Technology	Working Principle	Advantages	Limitations
RF Jamming	Disrupts communication signals	Long range	Affects nearby signals
Net Capture	Physically traps drone	Safe interception	Limited range
Vision-Based Tracking	Uses AI and computer vision	Low cost and accurate	Lighting dependent
Laser Systems	Directed energy attack	High precision	Very expensive
spark-actuated missile launcher	Electronic discharge response	Simple implementation	Short range

2.3 Artificial Intelligence in Surveillance

Artificial Intelligence has transformed modern surveillance systems by enabling automated monitoring and decision-making capabilities. AI-based systems can process large amounts of image and video data with high speed and accuracy.

Deep learning algorithms such as Convolutional Neural Networks (CNNs) are extensively used in image recognition, object classification, facial recognition, and autonomous navigation systems. These algorithms automatically learn features from training datasets and improve their performance over time.

In surveillance applications, AI is commonly used for:

- Object detection
- Human activity recognition
- Traffic monitoring
- Intrusion detection
- Facial recognition
- Intelligent tracking systems

The use of artificial intelligence in drone detection systems improves accuracy, reduces manual intervention, and enables real-time automated surveillance.

2.4 YOLO-Based Object Detection Systems

YOLO (You Only Look Once) is one of the most widely used deep learning algorithms for real-time object detection. It was first introduced by Joseph Redmon and has evolved through multiple versions including YOLOv3, YOLOv5, YOLOv7, and YOLOv8.

Traditional object detection methods perform classification and localization separately, making them computationally slow. YOLO performs object detection in a single stage by simultaneously predicting bounding boxes and class probabilities.

YOLOv8 offers several improvements over previous versions such as:

• Faster processing speed
• Better detection accuracy
• Improved small object detection
• Reduced computational complexity
• Real-time performance

Several research studies have demonstrated the successful use of YOLO algorithms in drone detection, vehicle detection, traffic monitoring, and security surveillance systems. Due to its efficiency and real-time capability, YOLOv8 has been selected for this project.

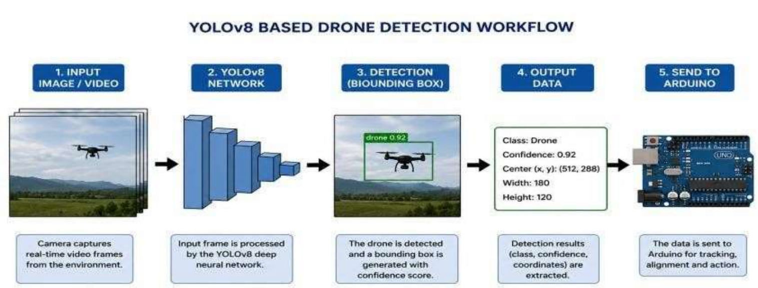

Figure 2.2: YOLOv8 based Drone Detection Workflow

2.5 Arduino-Based Embedded Systems

Arduino is an open-source microcontroller platform widely used in robotics, embedded systems, and automation applications. It provides a simple programming environment and supports interfacing with various sensors, motors, relays, communication modules, and electronic control devices.

Researchers have integrated Arduino with artificial intelligence systems for applications such as smart surveillance systems, autonomous robots, industrial automation, home automation, and intelligent tracking systems. Arduino-based systems are widely preferred because of their low cost, ease of programming, compact design, and compatibility with multiple hardware components.

In the proposed project, the Arduino Uno acts as the primary hardware controller responsible for real-time target tracking, SCAN mode, TRACK mode, TARGET LOCK verification, and spark-actuated missile launcher activation. The Arduino receives positional coordinates from the YOLOv8 computer vision system through serial communication and controls the movement of MG996R servo motors mounted on the pan-tilt mechanism using coordinate-based tracking commands.

The Arduino also controls the spark-generation mechanism used to actuate the missile launcher. When the target remains aligned near the center region of the image frame, the Arduino generates a FIRE command after TARGET LOCK verification and cooldown timing. The spark-actuated missile launcher module then produces a controlled spark demonstration under laboratory conditions.

The integration of Arduino with computer vision and artificial intelligence creates an intelligent embedded automation system capable of real-time drone detection, coordinate-based servo tracking, SCAN/TRACK operations, TARGET LOCK verification, FIRE command generation, and synchronized hardware-software operation.

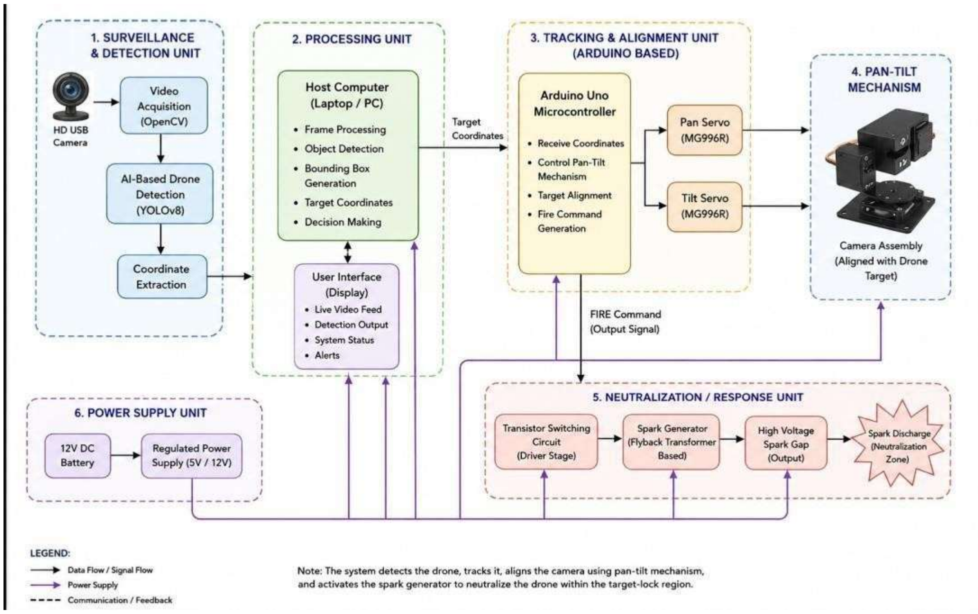

Figure 2.3: System Architecture of Anti Drone System

2.6 Existing Research Work

Several researchers have proposed intelligent drone detection and counter-UAV systems using different technologies such as radar systems, radio-frequency jamming, thermal imaging, net-capture mechanisms, interceptor drones, and artificial intelligence-based computer vision systems.

Radar-based drone detection systems are widely used for military and long-range surveillance applications. Although these systems provide accurate target detection capabilities, they require expensive hardware and complex signal processing techniques.

Some researchers have developed AI-based computer vision systems using deep learning algorithms such as YOLO, Faster R-CNN, and SSD for real-time drone detection and tracking. These systems achieved high detection accuracy and fast response performance using live video streams captured through cameras.

Several modern counter-UAV systems incorporate intelligent surveillance, automated tracking, and response mechanisms. These include RF monitoring systems, computer-vision-based tracking systems, embedded automation platforms, and experimental response technologies.

Research has also been conducted on Arduino-based embedded tracking systems where servo motors and pan-tilt mechanisms are controlled using positional coordinates generated through computer vision algorithms. Such systems demonstrate efficient target tracking capabilities at relatively low cost.

The literature review indicates that AI-based computer vision systems combined with embedded automation, coordinate-based tracking, and spark-actuated missile launcher mechanisms provide an effective solution for intelligent drone surveillance and real-time tracking applications.

2.7 Research Gaps

Although significant progress has been made in drone detection and counter-UAV technologies, several limitations still exist in currently available systems.

Most existing counter-drone systems are expensive and require specialized military-grade hardware such as radar systems, RF jammers, or advanced thermal imaging equipment. These systems are difficult to implement for low-cost educational or research purposes.

Many existing surveillance systems focus only on drone detection and do not integrate real-time target tracking and automated response mechanisms within a single platform. Some systems also suffer from limited real-time performance, complex hardware architecture, and high computational requirements.

There is limited research on low-cost AI-integrated drone detection and spark-actuated missile launcher systems using embedded automation and computer vision technologies. Existing systems often lack synchronized integration between object detection, tracking, target alignment, and automated electronic response activation.

Therefore, there is a need for an intelligent, low-cost, AI-based drone detection and tracking prototype capable of real-time surveillance, coordinate-based servo tracking, TARGET LOCK verification, and spark-actuated interception system activation using embedded automation technologies.

2.8 Preliminary Prototype Development

During the previous semester, an initial prototype of the AI-based drone detection and tracking system was successfully developed as the preliminary phase of the proposed project. The earlier work mainly focused on establishing the core architecture of the system, including drone detection, pan-tilt tracking, embedded control, and automated response mechanisms.

The preliminary prototype was developed using Arduino Uno, MG996R servo motors, HD USB camera, relay module, and a spark generation mechanism. The system utilized Python programming, OpenCV, YOLOv8 deep learning framework, and Arduino IDE for implementing real-time computer vision and embedded automation operations.

The initial development phase included the implementation of real-time YOLOv8-based drone detection using live video streams captured through the HD camera. Coordinate-based target tracking was performed using OpenCV, while the Arduino Uno controlled the pan-tilt mechanism for continuous target alignment. Serial communication was successfully established between the Python processing environment and the Arduino controller for synchronized operation of the hardware and software modules.

During the preliminary testing phase, several system functionalities were successfully demonstrated, including YOLOv8 drone detection, pan-tilt tracking, TARGET LOCK verification, Arduino integration, and spark-actuated response mechanism operation. The prototype demonstrated effective real-time performance under controlled laboratory conditions and confirmed the feasibility of integrating artificial intelligence, computer vision, embedded systems, and automation technologies into a unified surveillance platform.

The previous semester work provided the foundation for the complete system developed in the present thesis. Based on the observations and limitations identified during the preliminary phase, the current project introduces improved system organization, enhanced tracking methodology, optimized workflow integration, and more detailed implementation for intelligent drone surveillance and tracking applications.

2.9 Objectives of the Proposed Work

The primary objective of the proposed project is to develop an intelligent AI-based drone detection and tracking prototype capable of performing real-time surveillance, automated target tracking, and embedded control operations using computer vision and artificial intelligence technologies. The system is designed to provide an efficient, low-cost, and reliable solution for intelligent aerial surveillance applications under controlled laboratory conditions.

The specific objectives of the proposed work are as follows:

1. To develop a real-time drone detection system using the YOLOv8 deep learning algorithm for accurate identification of drones from live video streams.
2. To design and implement a servo-based pan-tilt tracking mechanism using Arduino Uno and MG996R servo motors for continuous target alignment and movement tracking.
3. To implement coordinate-based tracking and TARGET LOCK verification for maintaining accurate alignment between the detected drone and the tracking mechanism.
4. To implement an Arduino-controlled spark-actuated missile launching mechanism for demonstrating automated response activation under laboratory-scale testing conditions.
5. To establish serial communication between the Python-based AI processing system and the Arduino controller for synchronized hardware-software interaction.
6. To integrate artificial intelligence, computer vision, embedded systems, and automation technologies into a unified intelligent drone surveillance and tracking platform.
7. To evaluate the performance of the proposed system in terms of real-time detection capability, tracking stability, response speed, and overall system synchronization.

The proposed project aims to demonstrate the practical implementation of AI-integrated surveillance technologies using YOLOv8, OpenCV, Arduino Uno, and embedded automation systems for intelligent drone monitoring, tracking, and automated response applications under controlled laboratory conditions.

2.10 Proposed System

The proposed system combines artificial intelligence, computer vision, embedded systems, SCAN/TRACK operations, coordinate-based tracking, TARGET LOCK verification, and spark-actuated missile launcher activation to develop an intelligent drone detection and tracking prototype.

The system uses a high-definition camera to capture live video streams, which are processed using the YOLOv8 deep learning algorithm for real-time drone detection. Once the drone is detected, the system extracts positional coordinates and transmits them to the Arduino Uno microcontroller through serial communication.

The Arduino controls MG996R servo motors mounted on a pan-tilt mechanism to continuously align the tracking system with the moving target. A coordinate-based tracking mechanism is implemented where the Arduino continuously calculates positional error between the drone coordinates and image frame center coordinates.

Based on the error values, LEFT, RIGHT, UP, and DOWN commands are generated for servo alignment and TARGET LOCK verification.

An electronic spark-generation module is integrated as the prototype neutralization mechanism. When the drone enters the target alignment zone, the Arduino activates a digital output pin connected to the spark-generation module through a relay circuit. This produces control missile launch toward the target under laboratory conditions under laboratory conditions.

The proposed system provides real-time detection capability, intelligent target tracking, low-cost implementation, embedded automation, and synchronized hardware-software integration. The project demonstrates the practical application of artificial intelligence, computer vision, embedded automation, coordinate-based servo tracking, TARGET LOCK operations, and spark-actuated missile launcher activation in intelligent drone surveillance systems.

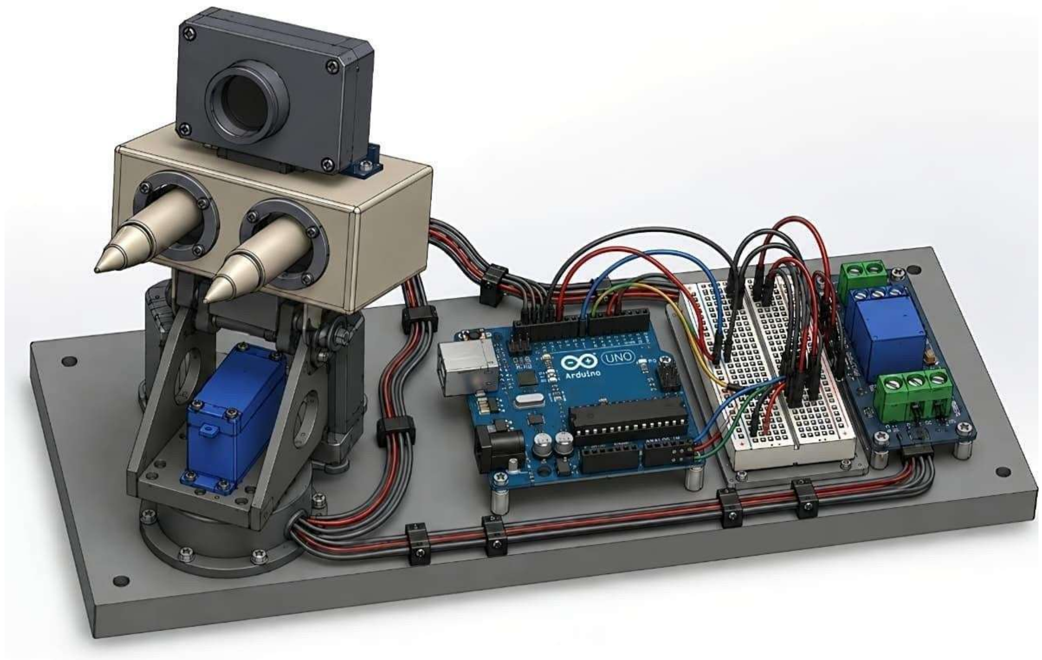

Figure 2.4: Assembled 3-D Model

2.11 Conclusion

This chapter reviewed the existing literature related to drone detection technologies, artificial intelligence-based surveillance systems, YOLO object detection algorithms, and Arduino-based embedded automation systems.

The review identified several limitations in existing systems including high implementation cost, lack of hardware integration, and limited real-time performance. The proposed project addresses these challenges by integrating YOLOv8, coordinate-based servo tracking, SCAN/TRACK operations, TARGET LOCK verification, and Arduino-controlled spark-actuated missile launcher activation for intelligent real-time drone surveillance applications.

The next chapter discusses the detailed system design, hardware components, software tools, and methodology used for implementing the proposed system.

Table 2.3 Comparison of Detection Methods

Detection Method	Accuracy	Cost	Complexity
Radar	High	Very High	High
RF Detection	Medium	Medium	Medium
Computer Vision	High	Low	Low
Thermal Imaging	High	Very High	High

CHAPTER 3
SYSTEM DESIGN

3.1 System Overview

The proposed system is an AI-based drone detection and tracking system designed for real-time surveillance and intelligent monitoring applications. The system integrates artificial intelligence, computer vision, embedded systems, and automation technologies to detect and track drones automatically under controlled laboratory conditions.

The system uses the YOLOv8 deep learning algorithm for drone detection, OpenCV for image processing, and Arduino Uno for servo motor control and embedded automation. The overall design focuses on low-cost implementation, real-time operation, and reliable target tracking performance.

The proposed system operates using a high-definition USB camera that continuously captures live video frames. The captured frames are processed by the YOLOv8 model implemented in Python for drone detection. After detecting the drone, the system extracts the target coordinates and sends them to the Arduino Uno through serial communication.

The Arduino controls the pan-tilt servo mechanism using MG996R servo motors to continuously align the camera with the moving drone target. The system demonstrates real-time surveillance, target tracking, and automated control operations.

3.2 Block Diagram of Proposed System

The block diagram of the proposed anti-drone system represents the flow of data, control signals, and automated operations between the hardware and software modules. The system consists of interconnected components working together for real-time drone detection, tracking, and electronic response activation.

The major components of the proposed system include:

1. HD Camera
2. YOLOv8 AI Detection Module
3. Python and OpenCV Processing System
4. Arduino Uno Microcontroller
5. Pan Servo Motor
6. Tilt Servo Motor
7. Pan-Tilt Tracking Assembly
8. Relay Module
9. Spark Generation Module
10. Power Supply Unit

The HD camera continuously captures real-time video frames from the surrounding environment. These video frames are transferred to the computer system where the YOLOv8 deep learning model processes the frames and identifies drone targets. Once detection occurs, the center coordinates of the drone are calculated.

The Python processing program sends the positional coordinates to the Arduino Uno using serial communication through the USB interface. The Arduino continuously receives updated coordinates and computes the required angular movement for the servo motors.

The pan servo motor rotates horizontally while the tilt servo motor rotates vertically. Together, these motors maintain the alignment of the camera and electronic neutralization assembly with the detected drone target.

When the target enters the predefined target-lock zone, the Arduino activates a digital output pin connected to a relay module. The relay safely switches the spark-generation circuit and activates the electronic response mechanism.

The block diagram demonstrates the synchronized interaction between artificial intelligence, computer vision, embedded control systems, and electronic automation mechanisms.

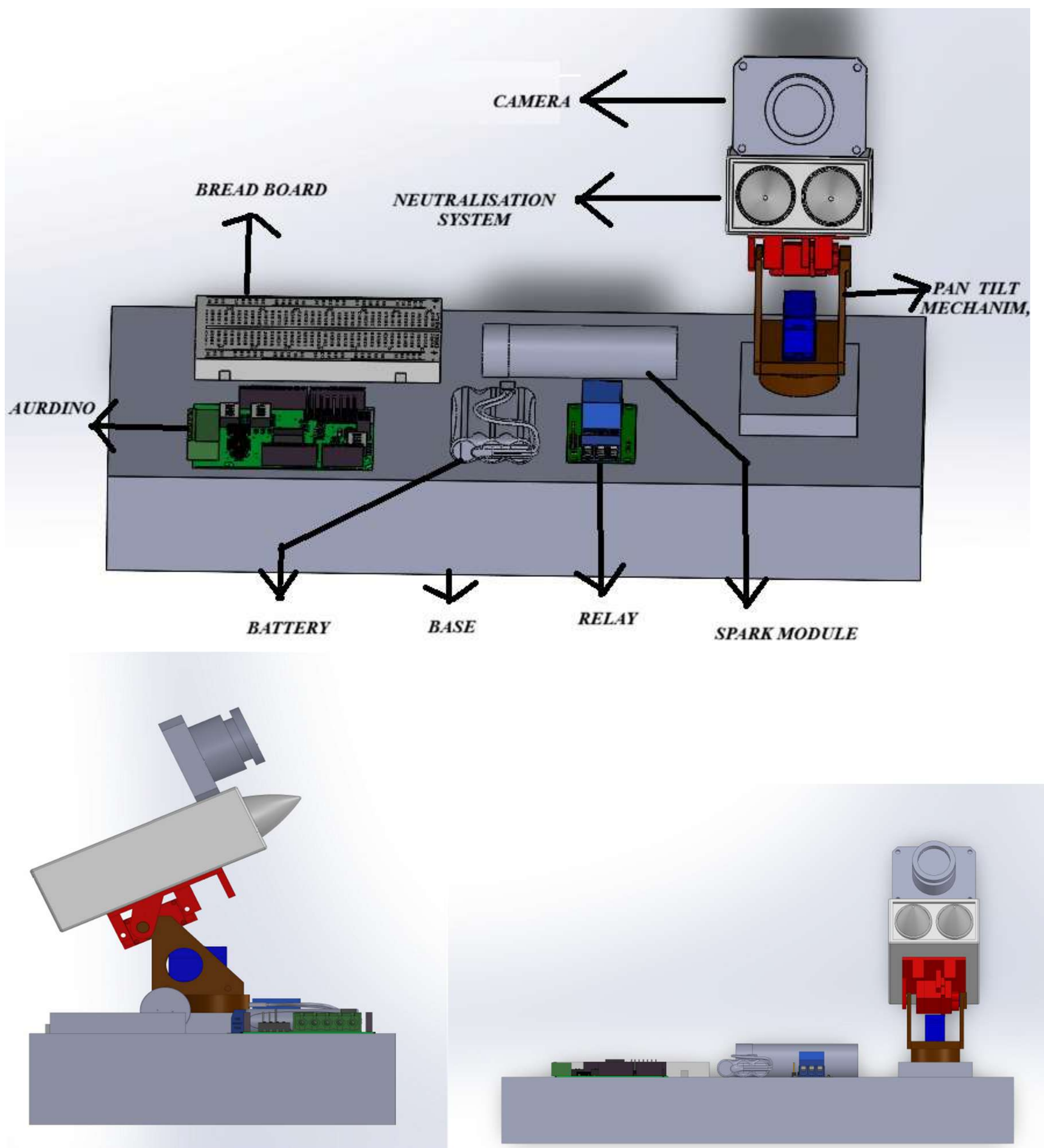

Figure 3.1: System Design and Architecture of the Proposed Anti-Drone System

3.3 Hardware Components and Selection

The hardware components used in the anti-drone system were selected based on cost efficiency, reliability, compatibility, response speed, and ease of integration with artificial intelligence and embedded automation systems.

HD USB Camera:

A high-definition USB webcam is used for capturing real-time video streams. The camera acts as the primary sensing device of the surveillance system. A 1080p resolution camera was selected to improve object detection accuracy and image clarity during real-time operation. The camera continuously sends live frames to the computer for YOLOv8 processing.

Arduino Uno:

The Arduino Uno board based on the ATmega328P microcontroller is used as the primary embedded control system. Arduino Uno was selected because of its low cost, easy programming environment, sufficient digital I/O pins, PWM support, and compatibility with servo motors and transistor-based switching circuits. The Arduino receives target coordinates from the computer and controls SCAN mode, TRACK mode, TARGET LOCK verification, servo movement, and FIRE command generation for spark-actuated missile launcher activation.

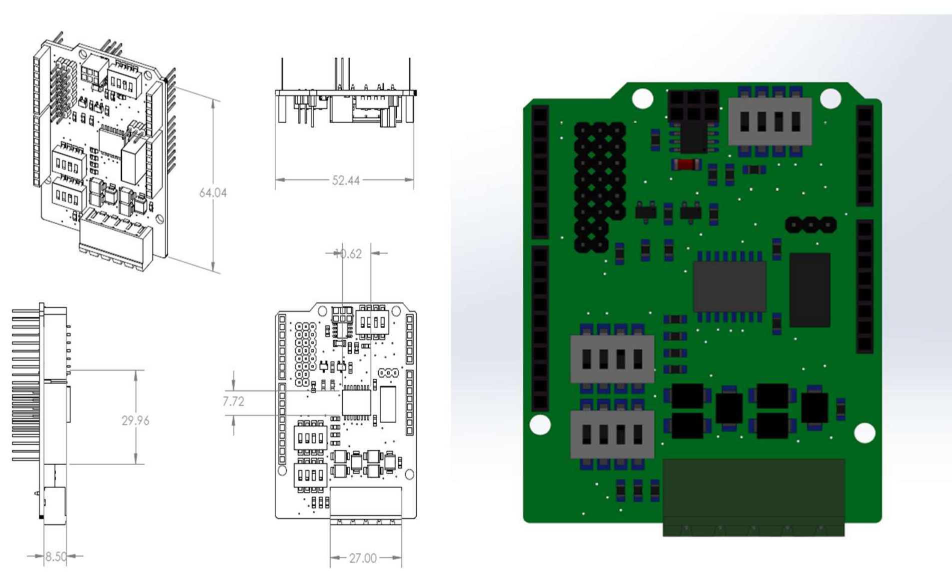

Figure 3.3 a : Sketch and Design of Arduino Uno

MG996R Servo Motors:

Two MG996R high-torque servo motors are used for implementing the pan-tilt tracking mechanism. These motors were selected because they provide fast angular response, high torque output, and accurate positional control. One servo controls horizontal movement while the other controls vertical movement.

Pan-Tilt Assembly:

The pan-tilt mechanism supports the camera and spark-actuated missile launcher module. The assembly enables coordinate-based target tracking in both horizontal and vertical directions using servo-controlled movement.

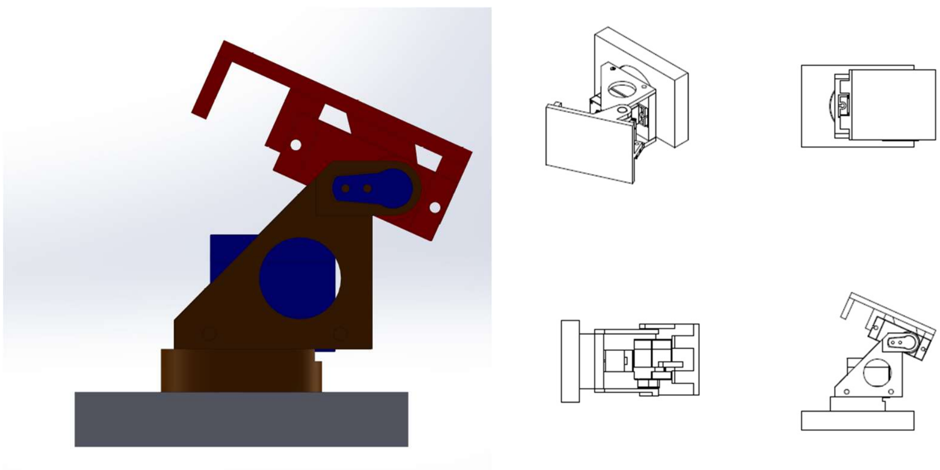

Figure 3.3 b : Sketch and Design of Pan-Tilt mechanism

Spark Generation Module:

The spark-generation system acts as the electronic neutralization prototype. The module generates controlled electrical sparks under laboratory conditions when activated through the relay system.

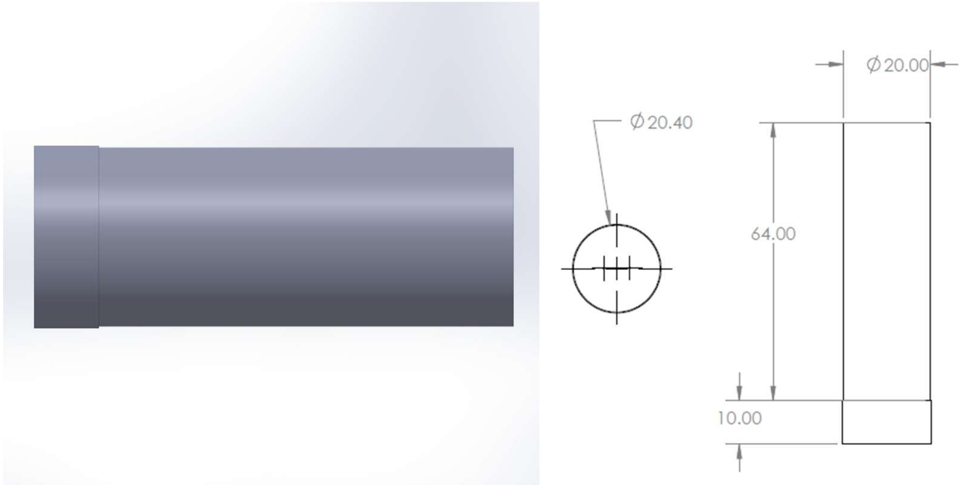

Figure 3.3 c : Design and Sketch of Spark Generator

Power Supply Unit:

A regulated power supply is used for providing stable operating voltage to the Arduino, servo motors, transistor switching circuit, and spark-actuated missile launcher module. Stable voltage is necessary for maintaining reliable operation of the tracking system.

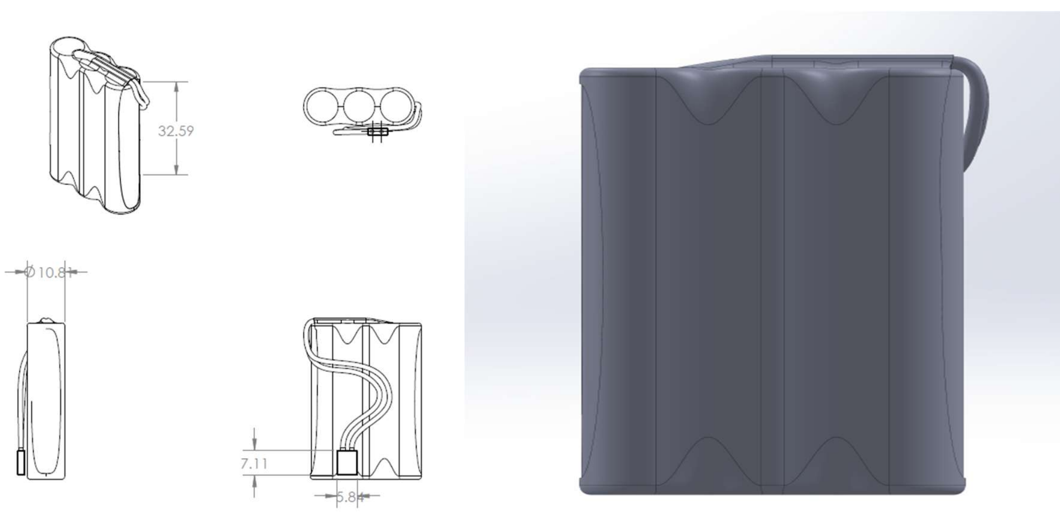

Figure 3.3 d : Sketch and Design of Power Supply

Table 3.1 Hardware Components Used in the Proposed System

Component	Specification	Purpose
Arduino Uno	ATmega328P	Embedded control and automation
HD USB Camera	1080p Webcam	Real-time drone surveillance
MG996R Servo Motors	High torque servo motors	Pan-tilt tracking
Relay Module	5V Relay	Electrical isolation and switching
Spark Generation Module	High voltage spark unit	Electronic response mechanism
Power Supply	5V / 12V DC	System power distribution

3.4 Software Requirements

The software environment of the proposed anti-drone system consists of artificial intelligence frameworks, computer vision libraries, programming environments, and embedded development tools required for implementing real-time drone detection and tracking operations.

Python Programming Language:

Python is used as the primary programming language because of its simplicity, flexibility, and extensive support for artificial intelligence and computer vision libraries. The complete drone detection and tracking logic is implemented using Python.

OpenCV Library:

OpenCV is an open-source computer vision library used for image processing and frame analysis operations. OpenCV is responsible for webcam interfacing, video frame acquisition, image display, coordinate extraction, and tracking visualization.

YOLOv8 Framework:

YOLOv8, developed by Ultralytics, is used as the deep learning model for real-time drone detection.The model processes image frames and identifies drones with high accuracy and fast inference speed.

Roboflow:

Roboflow is used for dataset preparation and image annotation. Drone images were collected and annotated using Roboflow tools before exporting them in YOLO format for training.

Jupyter Notebook:

Jupyter Notebook provides an interactive development environment for training, testing, debugging, and evaluating the YOLOv8 model.

Arduino IDE:

The Arduino Integrated Development Environment (IDE) is used for writing and uploading control programs to the Arduino Uno microcontroller. The Arduino program controls SCAN mode, TRACK mode, coordinate-based servo positioning, TARGET LOCK verification, and FIRE command generation for spark-actuated missile launcher activation.

Serial Communication:

Serial communication is implemented between the Python processing system and Arduino Uno for transmitting real-time target coordinates during tracking operations.

Table 3.2 Software Tools and Frameworks

Software / Framework	Purpose	Application
Python	Programming language	AI and computer vision
OpenCV	Image processing library	Frame processing and tracking
YOLOv8	Deep learning model	Drone detection
Roboflow	Dataset preparation	Image annotation
Arduino IDE	Embedded development	Arduino programming
Jupyter Notebook	Interactive environment	Model training and testing

3.5 Conclusion

This chapter discussed the complete system design of the proposed AI-based drone detection and tracking system. The chapter included a system overview, block diagram, hardware components, software requirements, and workflow design. The proposed system demonstrates the integration of artificial intelligence, computer vision, embedded systems, and automation technologies for intelligent surveillance and real-time drone tracking applications

CHAPTER 4
SYSTEM METHODOLOGY

4.1 Introduction

This chapter explains the methodology used for the implementation of the proposed AI-based drone detection and tracking system. The methodology includes drone detection using YOLOv8, real-time video processing, coordinate-based target tracking, servo motor control, and embedded automation operations. The complete system operates through the integration of artificial intelligence, computer vision, and Arduino-based control mechanisms.

4.2 Working Principle

The working principle of the proposed system is based on real-time drone detection and automated target tracking. The HD camera continuously captures live video frames from the environment. These frames are processed using the YOLOv8 deep learning model for drone identification.

After detecting the drone, the center coordinates of the target are calculated using OpenCV. The coordinate data is transmitted to the Arduino Uno through serial communication. Based on the positional error values, the Arduino controls the pan and tilt servo motors to continuously align the camera with the moving target.

YOLOv8 detection, embedded automation, coordinate-based tracking, SCAN/TRACK operations, TARGET LOCK verification, FIRE command generation, and spark-actuated missile launcher activation technologies.

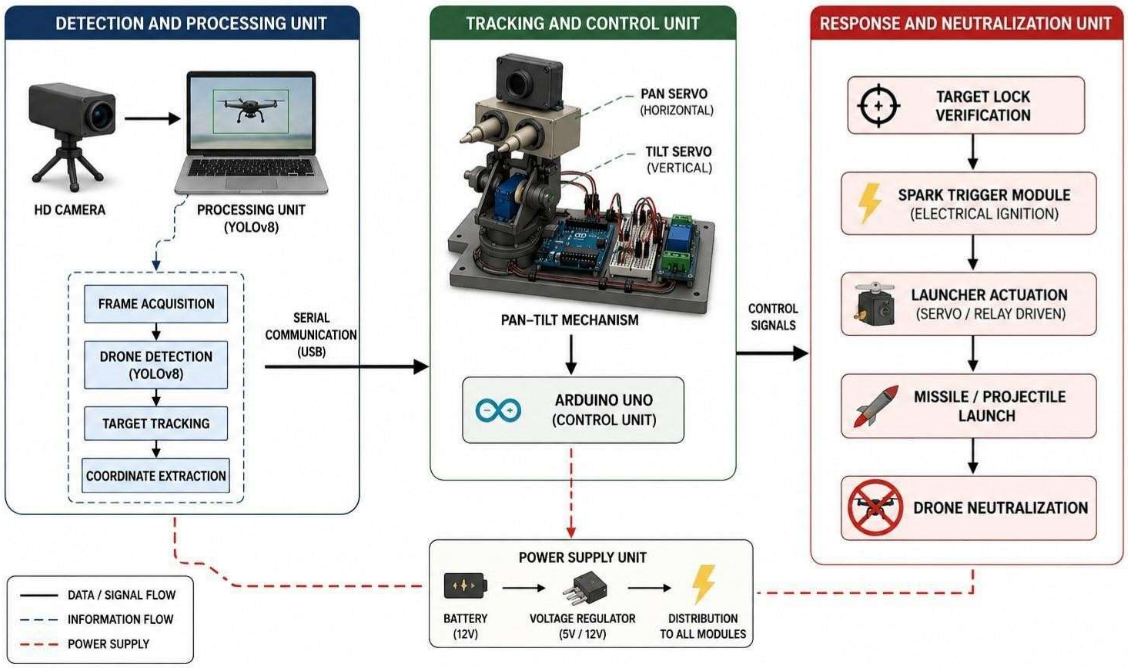

Figure 4.1: Working Principle of Proposed System

4.3 YOLOv8 Training Methodology

The YOLOv8 deep learning model is used for real-time drone detection. A drone image dataset was collected and annotated using Roboflow tools. The dataset was divided into training and validation sets before model training.

The YOLOv8 model was trained using Python in the Ultralytics framework. During training, the model learned to identify drones from different angles, sizes, and environmental conditions. After training completion, the best-performing model weights were used for real-time detection.

4.4 Pan-Tilt Tracking Mechanism

The pan-tilt mechanism is used for continuous drone tracking. Two MG996R servo motors are mounted to provide horizontal and vertical rotational movement.

The pan servo controls left and right movement, while the tilt servo controls upward and downward movement. The Arduino continuously adjusts the servo angles according to the target coordinates received from the Python processing system.

4.5 Electronic Neutralization Mechanism

The proposed system includes a relay-controlled spark generation mechanism for demonstrating automated response activation under laboratory conditions. When the target

aligns within the predefined tracking zone, the Arduino activates the relay module, which controls the spark generation circuit.
The mechanism demonstrates integration between AI-based detection and embedded automation systems.

4.6 Target Alignment and Tracking

Target alignment is performed using coordinate-based tracking. The center coordinates of the detected drone are compared with the center coordinates of the video frame.
If the target moves away from the center position, error values are generated. Based on these errors, movement commands such as LEFT, RIGHT, UP, and DOWN are sent to the servo motors for continuous target alignment and tracking.

4.7 Arduino Triggering Logic

Arduino Uno acts as the main embedded controller of the proposed system. The Arduino continuously receives target coordinates through serial communication and processes the tracking logic.
When the target enters the predefined alignment region, the Arduino activates the relay module for triggering operations. The triggering logic ensures synchronized operation between detection, tracking, and automated response mechanisms.

4.8 Safety Measures and Isolation

Safety measures were implemented to ensure stable and secure operation of the system. Electrical isolation was provided using the relay module to separate the high-voltage spark circuit from the Arduino controller.
Proper voltage regulation, controlled triggering conditions, and insulated wiring were used to reduce electrical risks during laboratory operation.

4.9 System Workflow

The system workflow begins with live video acquisition using the HD camera. The YOLOv8 model processes the frames and detects drones in real time. The detected target coordinates are transmitted to the Arduino Uno for servo-based tracking operations.

The pan-tilt mechanism continuously aligns the camera with the target. When the target satisfies the alignment condition, the relay-controlled spark generation mechanism is activated under controlled laboratory conditions.

4.10 Conclusion

This chapter explained the methodology used in the proposed AI-based drone detection and tracking system. The chapter discussed the working principle, YOLOv8 training methodology, pan-tilt tracking mechanism, target alignment process, Arduino triggering logic, safety measures, and complete operational workflow. The proposed methodology demonstrates the integration of artificial intelligence, computer vision, embedded systems, and automation technologies for intelligent drone surveillance applications.

CHAPTER 5
IMPLEMENTATION AND RESULTS

5.1 Introduction

This chapter presents the implementation details, experimental setup, testing procedures, and performance evaluation of the proposed AI-Based Vision-Guided Drone Detection, Tracking, and spark-actuated missile launcher Prototype using YOLOv8 and Arduino.

The chapter explains the practical implementation of hardware and software components integrated within the anti-drone system. The complete system was tested under controlled laboratory conditions using real-time drone detection and tracking operations. The chapter also discusses model training results, servo tracking performance, TARGET LOCK verification, spark-actuated missile launcher testing, and overall system performance evaluation.

The implementation stage demonstrates the successful integration of artificial intelligence, computer vision, embedded systems, coordinate-based tracking, and automated spark-actuated missle launcher technologies for intelligent surveillance applications.

5.2 Hardware Implementation

The hardware implementation of the proposed anti-drone system includes the integration of the HD camera module, Arduino Uno controller, MG996R servo motors, pan-tilt tracking mechanism, transistor-based switching circuit, spark-actuated missile launcher module, and regulated power supply unit.

The HD USB camera was mounted on the pan-tilt assembly for continuous monitoring and real-time drone tracking. The camera continuously captured video frames and transmitted them to the computer system for YOLOv8 processing.

The Arduino Uno was connected to the computer using serial communication through a USB cable. Two MG996R servo motors were connected to the PWM pins of the

Arduino controller for horizontal and vertical target alignment.

The transistor switching circuit was connected between the Arduino and spark-actuated missile launcher module for controlled spark activation. The spark-actuated missile launcher module was mounted on the tracking assembly to demonstrate laboratory-scale spark-actuated missile launcher operation during TARGET LOCK conditions.

The complete hardware assembly was securely mounted to reduce vibration and improve tracking stability during testing operations.

5.3 Software Implementation

The software implementation of the proposed system was performed using Python programming language, OpenCV library, YOLOv8 deep learning framework, Roboflow dataset tools, Jupyter Notebook environment, and Arduino IDE.

The YOLOv8 object detection model was trained using a custom drone image dataset prepared through Roboflow. The trained model was integrated into the Python environment for real-time webcam-based detection.

OpenCV was used for:

1. Video frame acquisition
2. Image processing
3. Bounding box visualization
4. Coordinate extraction
5. Real-time display operations

The Python program continuously processed live video frames and extracted drone coordinates. These coordinates were transmitted to the Arduino Uno through serial communication.

The Arduino program controlled:

- Servo motor movement
- Pan-tilt alignment

• TARGET LOCK verification
• FIRE command generation
• spark-actuated missile launcher activation

The integration of Python and Arduino enabled synchronized communication between the artificial intelligence system and the embedded control hardware.

5.4 YOLOv8 Model Training and Testing

The YOLOv8 model was trained using a custom drone dataset collected and annotated through Roboflow. The dataset contained drone images captured under different environmental conditions, viewing angles, backgrounds, and lighting variations.

The training process involved image annotation, dataset splitting, data augmentation, model training, and validation testing. During training, the model learned to identify drones using bounding box localization and confidence-based classification.

The training parameters used during model training included:

5.4.1 Epochs: 50–100

5.4.2 Batch Size: 16

5.4.3 Image Size: 640 × 640

5.4.4 Confidence Threshold: 0.5

The trained model achieved stable detection accuracy during testing operations. Real-time webcam testing confirmed that the system was capable of detecting drones with reliable confidence scores and stable bounding box generation.

The model successfully identified drones under different movement conditions and maintained reliable real-time performance during continuous surveillance operations.

5.5 Real-Time Drone Detection Results

Real-time drone detection was tested using live webcam feeds under indoor laboratory conditions. The camera continuously captured video frames while the YOLOv8 model processed the frames in real time.

The model generated:

5.5.1 Bounding boxes

5.5.2 Object labels

5.5.3 Confidence scores

5.5.4 Center coordinates

Table 5.1 Drone Detection Performance

Parameter	Result
Average Confidence Score	0.82 – 0.95
Detection Accuracy	High
Bounding Box Stability	Stable
False Detection Rate	Low
Real-Time Detection	Successful

The drone was successfully detected at different positions within the camera frame. Detection performance remained stable under varying movement speeds and moderate lighting changes.

The real-time detection system demonstrated:

1. Fast object recognition
2. Accurate target localization
3. Stable confidence scores
4. Continuous frame processing
5. Real-time visualization

Table 5.2 Frame Processing Performance

Parameter	Observation
Average FPS	18 – 25 FPS
Detection Delay	< 1 second
Processing Speed	Fast
Video Feed Stability	Continuous
Real-Time Processing	Successful

The YOLOv8 model provided efficient detection performance suitable for intelligent surveillance and automated tracking applications.

5.6 Servo-Based Tracking Performance

The servo-based tracking system was tested to evaluate its ability to maintain alignment with the moving drone target.

The Arduino continuously received coordinate data from the Python processing system and updated servo motor positions accordingly. The pan servo controlled horizontal movement while the tilt servo controlled vertical alignment.

The tracking mechanism demonstrated:

5.6.1 Smooth rotational movement

5.6.2 Fast response speed

5.6.3 Stable target alignment

5.6.4 Continuous tracking capability

5.6.5 Reduced positional error

Table 5.3 Servo Tracking Performance

Parameter	Result
Horizontal Tracking	Stable
Vertical Tracking	Stable
Servo Response Speed	Fast
Tracking Continuity	Continuous
Positional Error	Reduced

The coordinate-based tracking logic minimized alignment error between the image center and drone position. The system successfully maintained the drone within the TARGET LOCK region during movement using LEFT, RIGHT, UP, and DOWN servo commands.

The MG996R servo motors provided sufficient torque and angular response for real-time tracking operations.

5.7 Target Alignment and Locking Results

Target alignment performance was evaluated by analyzing the positional accuracy of the pan-tilt mechanism during tracking operations.

Table 5.4 TARGET LOCK Evaluation

Parameter	Result
Lock Acquisition Time	1 – 2 seconds
Target Stability	Maintained
TARGET LOCK Verification	Successful
Alignment Threshold	Maintained
Continuous Lock Duration	Stable

The system continuously calculated the tracking error between:

5.7.1 Image center coordinates

5.7.2 Drone center coordinates

The Arduino adjusted servo positions to reduce the tracking error and maintain target alignment.

The TARGET LOCK condition was achieved when:

1. Positional error remained within threshold limits
2. Target remained stable within alignment zone
3. Servo movement stabilized

The system successfully maintained TARGET LOCK during continuous drone movement within the camera field of view.

The TARGET LOCK mechanism improved synchronization between the detection system, servo tracking system, and spark-actuated missile launcher activation mechanism.

5.8 spark-actuated missile launcher Testing

The spark-actuated missile launcher mechanism was tested under controlled laboratory conditions using the Arduino-triggered spark-actuated missile launcher system.

Table 5.5 spark-actuated missile launcher Testing Results

Parameter	Result
FIRE Command Generation	Successful
Spark Activation	Successful
Trigger Reliability	High
Activation Delay	Minimal
Synchronization with TARGET LOCK	Stable

The spark-actuated missile launcher module was connected to the Arduino controller through a transistor-based switching circuit. Once the drone remained aligned near the center region of the image frame, the Arduino verified TARGET LOCK conditions and generated FIRE commands to activate the spark-actuated missile launcher circuit.

The testing process demonstrated:

1. Successful transistor-based triggering

2. Reliable FIRE command execution
3. Controlled spark-actuated missile launcher activation
4. Fast response time
5. Stable synchronization with TARGET LOCK mechanism

The spark-actuated missile launcher mechanism successfully demonstrated synchronized integration between artificial intelligence-based surveillance systems, embedded automation technologies, and TARGET LOCK-based activation control.

5.9 System Performance Evaluation

The overall system performance was evaluated based on detection accuracy, tracking stability, response time, synchronization, and hardware reliability.

The YOLOv8 model achieved reliable real-time drone detection with stable confidence scores. The servo-based tracking system demonstrated accurate target alignment and smooth rotational movement.

The Arduino controller maintained stable communication between the computer vision system and hardware modules. The transistor-switched spark-actuated missile launcher mechanism successfully activated during TARGET LOCK conditions.

Table 5.6 Arduino Communication Results

Parameter	Observation
Serial Communication	Stable
Coordinate Transmission	Continuous
Python-Arduino Synchronization	Successful
Data Transfer Delay	Minimal
Embedded Control Response	Reliable

Table 5.7 Environmental Testing Conditions

Parameter	Observation
Lighting Condition	Moderate Indoor Lighting
Camera Distance	1 – 5 meters
Drone Motion	Slow to Moderate
Testing Environment	Controlled Laboratory
Background Complexity	Simple Indoor Background

Table 5.8 System Stability Evaluation

Parameter	Result
Runtime Stability	Stable
System Crashes	Not Observed
Hardware Heating	Minimal
Tracking Interruptions	Rare
Overall Stability	Reliable

The system demonstrated:

5.9.1 Real-time operation

5.9.2 Low-cost implementation

5.9.3 AI-integrated automation

5.9.4 Efficient target tracking

5.9.5 Reliable spark-actuated missile launcher triggering

5.9.6 Stable hardware-software synchronization

Table 5.9 Overall System Performance Summary

Feature	Result
YOLOv8 Detection	Successful
Real-Time Processing	Achieved
Servo Tracking	Stable
TARGET LOCK	Verified
Spark Response	Successful
Arduino Communication	Stable
Hardware Integration	Successful

The overall prototype successfully achieved the objectives of intelligent drone detection, coordinate-based tracking, TARGET LOCK verification, and automated spark-actuated missile launcher activation.

5.10 Advantages and Limitations

The proposed anti-drone system provides several advantages including real-time drone detection, low-cost hardware implementation, intelligent servo tracking, TARGET LOCK verification, automated spark-actuated missile launcher activation, and easy integration between AI and embedded systems.

Major advantages include:

1. Real-time AI-based detection
2. Automated target tracking
3. Low-cost implementation
4. Embedded automation integration
5. Simple hardware architecture
6. Real-time visualization

However, the system also has certain limitations:

1. Limited operating range
2. Dependence on camera visibility
3. Sensitivity to lighting conditions
4. Limited outdoor testing

5. Small-scale laboratory implementation

Future improvements may include advanced cameras, improved tracking algorithms, thermal imaging integration, enhanced coordinate-based tracking, and improved autonomous response mechanisms.

5.11 Conclusion

This chapter presented the implementation details, experimental setup, testing procedures, and performance evaluation of the proposed anti-drone system.

The chapter discussed the hardware assembly, software implementation, YOLOv8 training process, real-time drone detection, servo-based tracking mechanism, TARGET LOCK system, and spark-actuated missile launcher testing.

Experimental testing confirmed the successful integration of artificial intelligence, computer vision, embedded systems, coordinate-based tracking, and automated spark-actuated missile launcher technologies. The proposed system demonstrated efficient real-time drone detection, stable tracking performance, TARGET LOCK verification, and synchronized spark-actuated missile launcher activation under controlled laboratory conditions.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Introduction

This chapter presents the conclusion and future scope of the proposed AI-Based Vision-Guided Drone Detection, Tracking, and spark-actuated missile launcher Prototype using YOLOv8 and Arduino. The chapter summarizes the overall achievements of the project, discusses the effectiveness of the developed anti-drone system, identifies system limitations, and highlights future improvements for enhancing system performance and scalability.

The proposed system successfully demonstrated the integration of artificial intelligence, computer vision, embedded systems, servo-based automation, TARGET LOCK verification, and spark-actuated missile launcher technologies into a single intelligent surveillance prototype. Experimental testing confirmed the capability of the system to perform real-time drone detection, automated target tracking, TARGET LOCK verification, and synchronized spark-actuated missile launcher activation under controlled laboratory conditions.

The project demonstrated the practical implementation of computer vision, machine learning, embedded automation, and intelligent surveillance technologies using low-cost and easily available hardware components.

6.2 Conclusion

The primary objective of this project was to develop a low-cost intelligent anti-drone prototype capable of real-time drone detection, coordinate-based tracking, TARGET LOCK verification, and automated spark-actuated missile launcher activation using artificial intelligence and embedded automation technologies.

Table 6.1 Experimental Outcome Summary

Test Parameter	Observation
Drone Detection	Successful
Coordinate Extraction	Accurate
Target Tracking	Stable
Servo Alignment	Smooth
spark-actuated missile launcher Activation	Reliable

The proposed system successfully utilized the YOLOv8 deep learning algorithm for accurate drone detection using live webcam video streams. The trained model demonstrated stable detection performance and reliable target localization under different testing conditions. The use of artificial intelligence significantly improved the automation capability and real-time surveillance performance of the system.

The Arduino Uno microcontroller successfully controlled the pan-tilt tracking mechanism using MG996R servo motors. The tracking system continuously adjusted the orientation of the camera assembly to maintain alignment with the moving drone target. The coordinate-based tracking mechanism reduced positional error and improved TARGET LOCK stability.

The spark-actuated missile launcher mechanism integrated with the transistor-switched spark-actuated missile launcher module demonstrated successful synchronization between computer vision, servo tracking, TARGET LOCK verification, and automated spark-actuated missile launcher activation. The system successfully triggered the spark-actuated missile launcher mechanism once the drone entered the predefined TARGET LOCK region.

The project demonstrated the practical implementation of:

1. Artificial intelligence-based surveillance
2. Real-time computer vision systems
3. Embedded automation technologies
4. Servo-based target tracking
5. spark-actuated missile launcher mechanisms

6. AI-integrated surveillance systems

The complete prototype achieved the objectives of intelligent drone detection, automated tracking, target alignment, TARGET LOCK verification, and controlled spark-actuated missile launcher activation under laboratory conditions. The project also demonstrated the interdisciplinary integration of computer vision, machine learning, embedded systems, and intelligent automation technologies in modern surveillance applications.

6.3 Major Achievements of the Project

The proposed anti-drone system achieved several important technical and functional objectives during implementation and testing.

Table 6.2 Major Achievements of the Project

Achievement	Status
YOLOv8 Detection	Implemented
Pan-Tilt Tracking	Successful
TARGET LOCK System	Verified
spark-actuated missile launcher Mechanism	Operational
Arduino Integration	Successful

The project successfully demonstrated how artificial intelligence and embedded systems can be integrated to develop intelligent anti-drone surveillance prototypes for educational and research applications.

6.4 Limitations of the System

Although the proposed anti-drone system demonstrated successful operation under laboratory conditions, certain limitations were observed during testing and implementation.

The major limitations of the system include:

1. Limited operational range due to webcam-based detection.

2. Dependence on lighting conditions and camera visibility.
3. Limited outdoor testing capability.
4. Small-scale laboratory implementation.
5. Limited tracking accuracy at very high drone speeds.
6. Dependence on stable serial communication.
7. Restricted spark-actuated missile launcher operating range.
8. Lack of autonomous navigation capability.
9. Limited environmental adaptability.
10. Absence of advanced sensor fusion mechanisms.

The system was developed mainly as a proof-of-concept educational prototype and not as a large-scale industrial surveillance or commercial counter-UAV system.

6.5 Future Scope

The proposed anti-drone system provides a strong foundation for future research and advanced development in intelligent surveillance and counter-UAV technologies. Several improvements and advanced features can be implemented in future versions of the system.

Future improvements may include:
1. Integration of thermal imaging cameras for night-time operation.
2. Use of high-resolution long-range cameras for extended detection distance.
3. Integration of autonomous navigation systems.
4. Implementation of advanced trajectory prediction algorithms.
5. Multi-drone detection and tracking capability.
6. Integration of radar and sensor fusion systems.
7. Use of more powerful embedded processors such as NVIDIA Jetson Nano or Raspberry Pi.
8. Wireless communication and IoT-based remote monitoring.
9. Advanced AI models for improved small-object detection.
10. Development of advanced autonomous surveillance platforms.

Future versions of the system may also incorporate advanced safety mechanisms, improved tracking stability, and intelligent decision-making algorithms for real-world surveillance applications.

The proposed research contributes to the growing field of AI-based surveillance systems and demonstrates the practical potential of artificial intelligence, computer vision, and embedded automation technologies in intelligent anti-drone applications.

6.6 Final Remarks

The proposed AI-Based Vision-Guided Drone Detection, Tracking, and spark-actuated missile launcher Prototype successfully demonstrated the integration of artificial intelligence, computer vision, embedded control systems, TARGET LOCK verification, and spark-actuated missile launcher technologies into a unified surveillance platform.

The project highlighted the importance of intelligent surveillance systems in addressing modern drone-related monitoring challenges. The successful implementation of real-time detection, automated tracking, TARGET LOCK verification, and spark-actuated missile launcher mechanisms demonstrates the practical applicability of AI-integrated surveillance systems in future security and monitoring applications.

The project also provided valuable practical experience in artificial intelligence, machine learning, computer vision, Arduino programming, embedded systems, and automation technologies, thereby contributing significantly to engineering research and educational learning.

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SOURCE CODE IMPLEMENTATION

A.1 Python Environment Setup Code

```
import subprocess
import sys
import os

def install(package):
    subprocess.check_call([
        sys.executable,
        "-m",
        "pip",
        "install",
        package
    ])

libraries = [
    "ultralytics",
    "opencv-python",
    "pyserial",
    "numpy",
    "matplotlib"
]

for lib in libraries:
    install(lib)

print("Libraries Installed Successfully")
```

A.2 YOLOv8 Model Training Code

```
from ultralytics import YOLO
import torch
```

```python
device = 0 if torch.cuda.is_available() else "cpu"

model = YOLO("yolov8n.pt")

results = model.train(
    data=YAML_PATH,
    epochs=10,
    imgsz=416,
    batch=4,
    workers=2,
    device=device,
    name="anti_drone_final",
    save=True
)

print("Training Completed")
```

A.3 Real-Time Drone Detection and Tracking Code

```python
from ultralytics import YOLO
import cv2
import serial
import time

arduino = serial.Serial('COM5', 115200)
time.sleep(2)

model = YOLO("best.pt")

cap = cv2.VideoCapture(0)

frame_center_x = 320
frame_center_y = 240

while True:

    ret, frame = cap.read()

    if not ret:
        break

    frame = cv2.resize(frame, (640, 480))
```

```
    results = model.predict(
        source=frame,
        conf=0.7,
        verbose=False
    )

    boxes = results[0].boxes

    for box in boxes:

        x1, y1, x2, y2 = map(int, box.xyxy[0])

        cx = (x1 + x2) // 2
        cy = (y1 + y2) // 2

        error_x = cx - frame_center_x
        error_y = cy - frame_center_y

        if error_x > 30:
            arduino.write(b"RIGHT\n")

        elif error_x < -30:
            arduino.write(b"LEFT\n")

        if error_y > 30:
            arduino.write(b"DOWN\n")

        elif error_y < -30:
            arduino.write(b"UP\n")

        if abs(error_x) < 20 and abs(error_y) < 20:
            arduino.write(b"FIRE\n")

    cv2.imshow("Anti-Drone System", frame)

    if cv2.waitKey(1) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()
```

A.4 Arduino Control Code

```
#include <Servo.h>

Servo panServo;
Servo tiltServo;

int panPos = 90;
int tiltPos = 90;

void setup() {

    Serial.begin(115200);

    panServo.attach(9);
    tiltServo.attach(10);

    pinMode(7, OUTPUT);
}

void loop() {

    if (Serial.available()) {

        String command =
            Serial.readStringUntil('\n');

        if (command == "LEFT") {
            panPos -= 2;
        }

        else if (command == "RIGHT") {
            panPos += 2;
        }

        else if (command == "UP") {
            tiltPos += 2;
        }

        else if (command == "DOWN") {
            tiltPos -= 2;
        }

        else if (command == "FIRE") {

            digitalWrite(7, HIGH);
```

```
            delay(500);
            digitalWrite(7, LOW);
        }

        panServo.write(panPos);
        tiltServo.write(tiltPos);
    }
}
```